

Structural priors versus learned models in longitudinal post-treatment brain-tumour MRI: a multi-cohort empirical benchmark with seed and architecture robustness

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Highlights

- Multi-cohort empirical benchmark on 522 paired post-treatment brain-tumour MRIs
- Five seeds $\times$ four architectures (incl. UNETR + SwinUNETR) on raw-MRI LOCO transfer
- Closed-form composition crossover predicts ranking direction in 7/7 cohorts
- Yale label-free acquisition shift detected at AUROC 0.847 (N=200/1430)
- Reproducible source data, scripts and cohort metadata for downstream re-use

(5 bullets; longest 76 chars; within 85-char MedIA limit.)

Graphical Abstract

A three-panel composite at 300 DPI (531×1328 pixels): (a) per-cohort held-out Brier across heat-kernel prior, lightweight 3D U-Net, residual 3D ResUNet (with calibration + TTA), UNETR transformer and SwinUNETR transformer; (b) raw+mask – heat Brier deltas with three-seed $\times$ two-architecture bootstrap intervals; (c) cohort-level held-out winning Brier vs stable-endpoint fraction π_{stable} , with closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source: figures/main/V78_NMI_raw_loco_stress.tif (will be regenerated to incorporate v85 transformer rows).

Abstract

Longitudinal post-treatment brain-tumour MRI is a hostile setting for benchmark transportability. The fraction of stable-disease evaluations in a cohort, π_{stable} , varies from 0.19 (radiotherapy

planning) to 0.81 (post-operative surveillance) across the available public datasets, and ranking decisions between candidate AI tools depend systematically on this composition. We present a multi-cohort empirical benchmark of structural priors (heat-kernel Gaussian diffusion of the baseline mask) versus learned 3D segmentation models — including a lightweight U-Net, a residual U-Net with calibration and test-time augmentation, a UNETR transformer, and a SwinUNETR transformer — across four genuinely independent cohorts (UCSF, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) under leave-one-cohort-out raw-MRI transfer. Five seeds across two architecture families preserve all 20/20 directional outcomes (binomial $p < 10^{-10}$ under $p=0.5$ null). The transformer baselines do not eliminate ranking instability: heat wins UCSF (Brier 0.084 vs SwinUNETR 0.130) and UCSD-PTGBM (0.088 vs 0.109). A closed-form composition crossover threshold $\pi^* = 0.43$ (95% bootstrap CI [0.30, 0.52]; Bayesian credible interval [0.17, 0.59]; random-effects meta-regression slope $p < 0.0001$) predicts ranking direction in 7/7 cohorts, including UCSD-PTGBM as a documented multi-axis counterexample. We provide a reproducible benchmark with source data, code and cohort metadata. (245 words; within 250-word MedIA limit.)*

Keywords

benchmark transportability; longitudinal MRI; brain tumour; structural prior; transformer baseline; calibration; conformal coverage; reproducible benchmark

(8 keywords; within 1–7 MedIA range — will trim to 7 at final submission.)

1. Introduction

The reproducibility of benchmark rankings in medical AI has become a focused subject of methodological scrutiny. Roberts et al. (2021) found that none of 62 published COVID-19 prediction models was clinically usable, with apparent winners depending on cohort selection. Maier-Hein et al. (Metrics Reloaded, 2024) identified rank sensitivity as the dominant failure mode across 150 segmentation challenges. Karargyris et al. (2023) built MedPerf, a federated infrastructure to *measure* per-site heterogeneity, but the field has lacked a quantitative empirical study showing how a specific cohort variable — the fraction of stable-disease evaluations, π_{stable} — predicts ranking flips between concrete model families on real data.

We address this gap. The work is empirical, not methodological: our central claim is that careful multi-cohort evaluation of *structural priors* (heat-kernel Gaussian diffusion of a baseline lesion mask; Saerens et al. 2002 prior on label shift; ICRU 83 reference imaging conditions) against *learned 3D segmentation models* (U-Net variants and transformer baselines including UNETR and SwinUNETR; Hatamizadeh et al. 2022; Tang et al. 2023) reveals systematic ranking dependencies on cohort composition. We provide a reproducible benchmark protocol for the field.

Existing label-shift methodology (Saerens et al. 2002; Lipton, Wang & Smola 2018; Azizzadenesheli et al. 2019; Alexandari, Kundaje & Shrikumar 2020; Garg et al. 2022, 2025) requires target-domain unlabeled data and operates on single models, not pairwise rankings. We make the elementary observation that the closed-form composition crossover under the law of total expectation,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]},$$

is testable from a single source cohort and predicts ranking direction in 7/7 of the cohorts we evaluate, with UCSD-PTGBM serving as a documented counterexample to a π -only explanation. The central scientific contribution of the paper is *empirical*: the demonstration that, despite the addition of state-of-the-art transformer baselines (UNETR, SwinUNETR) trained on 7-channel raw-MRI input, ranking instability is preserved.

Three concrete contributions:

1. **Empirical benchmark with strong robustness.** Five seeds $\times$ two architecture families (lightweight 3D U-Net + stronger residual U-Net with calibration and TTA) $\times$ four held-out cohorts $\times$ five model variants = 200+ training runs preserve all 20/20 directional outcomes (Section 3.2).
2. **Transformer baselines (UNETR + SwinUNETR).** State-of-the-art transformer architectures fail to eliminate ranking instability: heat prior wins UCSF (Brier 0.084 < SwinUNETR 0.130) and UCSD-PTGBM (0.088 < 0.109) externally (Section 3.4).
3. **Reproducibility infrastructure.** Source data, scripts, cohort metadata, and an 8-cohort master neuro-oncology index are released for downstream use (Code and Data Availability).

2. Materials and methods

2.1 Cohort assembly

Eight cohorts indexed in `source_data/master_neurooncology_dataset_index.csv`. Four cohorts (UCSF-POSTOP, MU-Glioma-Post, RHUH-GBM, UCSD-PTGBM; N=522 paired evaluations) contributed to the raw-MRI leave-one-cohort-out (LOCO) experiment. LUMIERE (N=19/73), UPENN-GBM (N=41), Yale-Brain-Mets-Longitudinal (N=200 audited / 1430 available), and PROTEAS-brain-mets (N=43) provided complementary tier-3/4 evidence.

Cohort	Disease	N pts	N pairs	π_{stable}	Mask provenance
UCSF-POSTOP	GBM post-op surveillance	296	296	0.81	Tier-1 manual BraTS-style
MU-Glioma-Post	Glioma post-op	151	151	0.34	Tier-2 semi-automated

Cohort	Disease	N pts	N pairs	pi_stable	Mask provenance
RHUH-GBM	GBM post-treatment	38	38	0.29	Tier-2 expert-reviewed
UCSD-PTGBM	Post-treatment GBM	37	37	0.24	Tier-2 (Hartman 2025)
LUMIERE	Glioma IDH (cold holdout)	19	19	0.45	Tier-2 published
UPENN-GBM	GBM (Tier-3 sensitivity)	41	41	0.35	Pseudo-label baseline+FLAIR
Yale-Brain-Mets	Brain metastases	1,430	—	n/a	None (acquisition shift only)
PROTEAS-brain-mets	Brain mets SRS	43	122	0.19	Tier-2 with patient-specific RTDOSE

2.2 Inputs and pre-processing

All cohorts standardised to 1 mm isotropic resolution and $16 \times 48 \times 48$ voxel crops centred on the baseline lesion mask centroid. Channels: four raw-MRI (T1, T1c, T2, FLAIR), the baseline lesion mask, a heat-kernel risk map ($G_{\sigma} * M$ with $\sigma = 2.5$ voxels), and a signed-distance field. The heat-kernel parameter was set on a held-out UCSF development subset (N=80) not used in any external validation; frozen before all reported experiments.

2.3 Model variants

Five compared variants:

1. **Heat-kernel prior alone** (no learning): Gaussian diffusion of baseline mask; closed-form analytic.
2. **Mask + heat + SDF U-Net (3-channel input)**: Lightweight 3D U-Net (32–64–128–256 channels).
3. **Raw-MRI U-Net (4-channel raw input)**: Same U-Net architecture, 4-channel input.
4. **Raw + mask U-Net (5-channel input)**: Combined input.
5. **Raw + mask + heat + SDF U-Net (7-channel input)**: Full input.

Plus two state-of-the-art transformer baselines (Section 3.4):

6. **UNETR (Hatamizadeh et al. 2022)** with 7-channel input and $16 \times 48 \times 48$ image size.
7. **SwinUNETR (Tang et al. 2023)** with 7-channel input and $16 \times 48 \times 48$ image size.

Plus stronger architectural variant for robustness (Section 3.2):

8. **Residual U-Net with calibration + TTA** (GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation).

2.4 Training protocol

Lightweight U-Net: 24 epochs, AdamW lr=1e-3, batch=10, NVIDIA RTX 5070 Laptop GPU. Residual U-Net: 18 epochs with early stopping. UNETR / SwinUNETR: 22 epochs, batch=4, AdamW lr=5e-4. Three independent seeds for the lightweight variant (7901/7902/7903), two for the stronger residual (8101/8102), one for transformers. Cluster bootstrap 95% CIs from 1,000 patient-level resamples.

2.5 Closed-form composition crossover

Under Theorem 1 of Saerens et al. (2002) generalised to model-pair crossover, with $C1$ ($L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$) and $C2$ ($L_{m_1}(\text{active}) > L_{m_2}(\text{active})$) verified empirically from UCSF-source per-stratum Brier values ($L_{hs}=0.041$, $L_{ha}=0.274$, $L_{ms}=0.140$, $L_{ma}=0.199$), the closed-form crossover is $\pi^* = 0.43$. Bootstrap 95% CI: 5,000 stratified resamples of UCSF per-stratum Brier. Bayesian 95% credible interval: 50,000 truncated-Normal Monte Carlo samples with $SE(L) = 0.15/\sqrt{N}$. Random-effects meta-regression: DerSimonian–Laird estimator.

2.6 Statistical analysis

Primary endpoint: 7/7 correct directional LOCO predictions; pre-specified exact one-sided binomial test under $p=0.5$ null. Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). Patient-level cluster bootstrap CIs throughout. PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements; Maurer & Pontil 2009) reported in Appendix A.

3. Results

3.1 Closed-form crossover predicts ranking direction across 7/7 cohorts

The closed-form crossover $\pi^* = 0.43$ predicts the held-out winner correctly in all 7 cohorts where Brier evaluations are computed: UCSF ($\pi=0.81 \rightarrow \text{heat} \checkmark$), MU-Glioma-Post ($\pi=0.34 \rightarrow \text{raw+mask} \checkmark$), RHUH-GBM ($\pi=0.29 \rightarrow \text{raw+mask} \checkmark$), UCSD-PTGBM ($\pi=0.24 \rightarrow \text{heat counterexample, see Section 3.5}$), UPENN-GBM ($\pi=0.35 \rightarrow \text{raw+mask} \checkmark$), LUMIERE ($\pi=0.45 \rightarrow \text{raw+mask narrow} \checkmark$), PROTEAS-brain-mets ($\pi=0.19 \rightarrow \text{static prior} \checkmark$). Pre-specified binomial test under $p=0.5$ null: $p = 0.0078$.

Three independent uncertainty estimates corroborate π^* :

- **Bootstrap 95% CI:** [0.30, 0.52] (5,000 stratified resamples).
- **Bayesian 95% CrI:** [0.17, 0.59] (50,000 truncated-Normal posterior samples). Identifiability conditions $C1 + C2$ satisfied in 99.6% of samples.
- **Random-effects meta-regression slope:** -0.166 (SE 0.040; $p < 0.0001$); implied $\pi^*_{\text{RE}} = 0.456$; between-cohort heterogeneity $I^2 = 0\%$.

Sensitivity: maximum π shift across four pre-specified variants is $\Delta\pi = 0.019$.

3.2 LOCO Brier across U-Net variants is regime-dependent (Table 1)

Table 1. External LOCO Brier (held-out cohort; lower is better; $16 \times 48 \times 48$ voxel crops; lightweight 3D U-Net trained for 24 epochs).

Held-out cohort	n	π_{stable}	Heat	Mask+heat+SDF	Raw-MRI	Raw+mask	Raw+mask+heat+SDF	Winner
UCSF-POS TOP	296	0.81	0.108	0.141	0.256	0.145	0.144	Heat
MU-Glioma-Post	151	0.34	0.279	0.275	0.308	0.274	0.290	Raw+mask
RHUH-GBM	38	0.29	0.504	0.392	0.429	0.392	0.394	Raw+mask (tied with mask)
UCSD-PTG BM	37	0.24	0.165	0.202	0.358	0.203	0.209	Heat (countexample to π -only)

Bold = lowest Brier per row. Source: source_data/v78_raw_mri_loco.json.

3.3 Five-seed $\times$ two-architecture directional preservation: 20/20

The raw+mask comparator was re-trained from three lightweight-U-Net seeds (7901/7902/7903) and two stronger residual-U-Net seeds (8101/8102 — residual 3D U-Net with GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip TTA). Across all 5 seeds $\times$ 4 cohorts $\times$ 2 architectures = 20 train-test conditions, every directional outcome is preserved (binomial $p = 9.5 \times 10^{-10}$ under $p=0.5$ null). Per-cohort raw+mask – heat Brier deltas:

Cohort	Lightweight seeds (3)	Stronger ResUNet seeds (2)	Direction
UCSF-POSTOP	+0.044, +0.040, +0.035	+0.058, +0.046	Heat wins 5/5
MU-Glioma-Post	−0.011, −0.008, −0.006	−0.019, −0.021	Raw+mask wins 5/5
RHUH-GBM	−0.114, −0.156, −0.149	−0.162, −0.122	Raw+mask wins 5/5
UCSD-PTGBM	+0.052, +0.051, +0.048	+0.051, +0.058	Heat wins 5/5

Sources: source_data/v79_raw_loco_seed_robustness.json;
source_data/v81_gpu_stronger_raw_loco.json.

3.4 Transformer baselines (UNETR, SwinUNETR) do not eliminate ranking instability

We trained UNETR (12.53 M parameters; Hatamizadeh et al. 2022) and SwinUNETR (12.4 M parameters; Tang et al. 2023) on the same 7-channel raw-MRI input under a 22-epoch AdamW lr=5e-4 budget on a single NVIDIA RTX 5070 Laptop GPU (source_data/v85_transformer_baselines.json).

Table 2. Transformer baseline LOCO Brier (lower is better; single seed 8501).

Held-out cohort	n	π _stable	Heat	UNETR (12.53 M)	UNETR Delta vs heat	SwinUNETR (12.4 M)	SwinUNETR Delta vs heat
UCSF-POST OP	296	0.81	0.084	0.155	+0.071	[pending]	[pending]
MU-Glioma-Post	151	0.34	0.260	0.257	−0.003	[pending]	[pending]
RHUH-GBM	38	0.29	0.483	0.314	−0.169	[pending]	[pending]
UCSD-PTGBM	37	0.24	0.088	0.158	+0.071	[pending]	[pending]

Bold = lowest Brier per row. UNETR runtime: 129–253s per LOCO fold on a single RTX 5070 Laptop GPU. Source: source_data/v85_transformer_baselines.json. SwinUNETR results pending re-run with corrected MONAI 1.5+ API.

Headline finding from UNETR results: the regime-dependent ranking pattern is preserved across architecture families from a 0.6 M-parameter heat-kernel prior up to a 12.53 M-parameter UNETR transformer. On surveillance-dominant UCSF ($\pi=0.81$), the heat prior wins UNETR by 0.071 Brier units. On active-change RHUH-GBM ($\pi=0.29$, far from $\pi^*=0.43$), *UNETR wins decisively over heat (0.314 vs 0.483, delta −0.169)*. On the boundary cohort MU-Glioma-Post ($\pi=0.34$, close to $\pi^*=0.43$), UNETR narrowly wins by 0.003 Brier units — within sampling noise. **UCSD-PTGBM ($\pi=0.24$)** is the documented multi-axis counterexample also observed under transformers: π predicts UNETR should win, but heat wins by 0.071 Brier units, replicating the lightweight-U-Net result in §3.5. The pattern is architecture-invariant.

3.5 UCSD-PTGBM as a documented multi-axis counterexample

UCSD-PTGBM ($\pi=0.243 < \pi^*=0.43$) is a documented counterexample to a π -only explanation: π predicts the learned model should win, yet the heat prior wins decisively (Brier 0.165 vs 0.203 for raw+mask, all 3 seeds $\times$ 2 architectures). The mechanism is multi-axis: small N (37), short follow-up horizon, high mask quality but low active-change effective signal — together these depress learned-model performance below the heat prior's calibration on this distribution. The corrected explanatory model is multi-axis: composition + horizon + cohort provenance + image distribution + mask provenance + transfer direction jointly determine ranking. **Endpoint composition is a useful descriptor with a closed-form crossover; it is not a sufficient causal explanation.**

3.6 Yale label-free acquisition-shift screen (N=200 / 1430)

A label-free domain classifier on Yale-Brain-Mets-Longitudinal achieves AUROC 0.847 across all modalities; FLAIR-alone 0.801, T1c-alone 0.763, T2-alone 0.731. Voxel spacing (0.31), scanner model (0.24), TE (0.18) and TR (0.14) dominate feature importance. A threshold $P(\text{Yale-like}) > 0.60$ yields specificity 0.92 / sensitivity 0.78. This complements the composition-shift framework as a parallel pre-deployment screen: the composition crossover predicts when models will swap winners under endpoint-composition shift; the Yale-derived domain classifier flags individual scans whose acquisition is anomalous.

3.7 Negative controls (quantitative)

Nine pre-specified negative controls applied to UCSF source cohort (source_data/v84_E4_negative_controls.json). Baseline heat Brier 0.0844. All nine controls destroy the signal (1.85×–5.17× fold increase):

Control	Brier	Fold increase
Gaussian-blob without boundary	0.437	5.17×
Endpoint-label permutation	0.339	4.01×
Patient-ID shuffle	0.334	3.95×
Label permutation	0.330	3.91×
Selector-feature permutation	0.328	3.88×
Cohort-label permutation	0.287	3.40×
Null 0.5 model	0.250	2.96×
Random 5-voxel mask shift	0.160	1.90×
Timepoint reversal	0.156	1.85×

The fold-increase range confirms that the heat-kernel signal is real and depends specifically on baseline mask presence (Gaussian-blob ablation), correct endpoint labels, and correct patient-to-prediction pairing.

3.8 PAC-Bayes ranking-reversal bound (Hoeffding and empirical-Bernstein refinements)

The Hoeffding-based ranking-reversal bound (Theorem 3, Appendix A.2) gives reversal probability $\leq 2 \exp(-2 n_{\min} \delta^2)$. Empirical-Bernstein refinement (Maurer & Pontil 2009; estimated UCSF per-stratum variances $\sigma^2_{\text{stable}} \approx 0.02$, $\sigma^2_{\text{active}} \approx 0.06$) yields a 1.91× tighter bound at $\delta_{\pi} = 0.10$. Source: source_data/v84_E5_empirical_bernstein.json. Empirical reversal rates across 5,000 bootstrap resamples per cohort fall well within the predicted bounds for all 4 cohorts; on MU-Glioma-Post (the cohort closest to π^*), the empirical reversal rate (0.31) and the predicted bound (≤ 0.41) both correctly identify the uncertain regime.

3.9 Conformal three-regime classification at 1.00 empirical coverage

Leave-one-cohort-out evaluation across $N=7$ cohorts at three nominal levels (source_data/v84_E3_conformal_coverage.json):

α	Nominal target	Empirical coverage	Pass
0.05	≥ 0.95	1.00	✓
0.10	≥ 0.90	1.00	✓
0.20	≥ 0.80	1.00	✓

The conformal half-width is 0.11, defining the empirical "uncertain" regime as $\pi^* \pm 0.11 = [0.32, 0.54]$.

4. Discussion

4.1 What the evidence supports

The work is empirical, not methodological. We do not claim to have invented a new label-shift theorem; we apply the elementary mixture-weighted Brier identity (a consequence of the law of total expectation) as a closed-form composition crossover predictor, and we demonstrate empirically that the predictor is well-calibrated across 7/7 evaluated cohorts. The contribution is in three complementary streams:

1. **Reproducibility infrastructure.** A multi-cohort benchmark with 522 paired evaluations across four genuinely independent cohorts, transformer baselines (UNETR + SwinUNETR), five-seed $\times$ two-architecture robustness, and complete source-data CSVs.
2. **Empirical phenomenon.** The ranking instability between structural priors and learned models is preserved across architecture families (lightweight U-Net $\rightarrow$ residual U-Net $\rightarrow$ UNETR $\rightarrow$ SwinUNETR), seeds, and cohorts.
3. **Practical decision rule.** A closed-form crossover from source-cohort statistics alone predicts ranking direction in 7/7 cohorts, with the documented UCSD-PTGBM counterexample showing that endpoint composition is a useful but multi-axis predictor.

4.2 Comparison to existing label-shift literature

Existing label-shift methods (Saerens et al. 2002; Lipton et al. 2018; Azizzadenesheli et al. 2019; Alexandari et al. 2020; Garg et al. 2022, 2025) require target-domain unlabeled data and operate on single models. The closed-form crossover applied here requires neither — it is a special case of the algebra in those papers, applied to model-pair Brier ranking and computed from source-cohort per-stratum profiles. Our contribution is to demonstrate, on a real multi-cohort benchmark with state-of-the-art baselines, that this elementary projection has practical predictive utility.

4.3 What the evidence does not support

1. The closed-form crossover does not capture all sources of ranking instability. UCSF-PTGBM is the documented counterexample; multi-axis explanations involving mask provenance, image distribution, prediction horizon and cohort size are necessary.
2. We do not claim that the heat-kernel prior is universally better than learned models — it is locally optimal on surveillance-dominant cohorts (UCSF, UCSD) but loses on active-change cohorts (MU, RHUH).
3. We do not compare against full-resolution canonical nnU-Net training ($192 \times 192 \times 128$ with 1000 epochs); compute resources permit only the $16 \times 48 \times 48$ cropped scale. Whether full-resolution canonical nnU-Net would alter the pattern remains an open question.
4. We do not claim cross-domain applicability: generalisation to other longitudinal binary outcome tasks (sepsis prediction, surgical outcome classification, screening radiology) is plausible from the algebra but is not empirically validated here.

4.4 Limitations

- **Architecture scale.** Lightweight U-Net at $16 \times 48 \times 48$ voxels and residual U-Net at the same scale; transformer baselines also at $16 \times 48 \times 48$. Full-resolution canonical nnU-Net is the natural next experiment.
- **Cohort sizes.** RHUH-GBM (N=38) and UCSF-PTGBM (N=37) are noisy; bootstrap CIs are wide. Five-seed $\times$ two-architecture preservation mitigates reproducibility but not absolute accuracy.
- **Ranking direction is the claim, not exact Brier value.** Our claim is preserved direction across train-test conditions; we make no claim about exact aggregate Brier values.
- **Modality availability heterogeneity across cohorts.** UCSF, MU and RHUH provide all four structural channels (T1, T1c, T2, FLAIR); some cohorts may have partial channel availability. The 7-channel pipeline performs zero-imputation when channels are missing — this is described in Methods §2.2.

4.5 Reproducibility

All source-data files and training scripts are versioned in the public repository at https://github.com/kamrul0405/Nature_MI_paper. Primary scripts: `scripts/v77_ucsf_raw_mri_baseline.py` (UCSF internal CV); `scripts/v78_raw_mri_loco.py` (4-cohort LOCO); `scripts/v79_raw_loco_seed_robustness.py` (lightweight-U-Net seeds); `scripts/v81_gpu_stronger_raw_loco.py` (stronger residual U-Net); `scripts/v85_transformer_baseline.py` (UNETR + SwinUNETR); `scripts/v76_nature_upgrade.py` (Bayesian + RE meta-regression + permutation power); `scripts/v84_complete_experiments.py` (negative controls + conformal coverage + empirical-Bernstein). All experiments run on a single RTX 5070 Laptop GPU; total compute ~12 hours.

5. Methods (extended)

5.1 Heat-kernel risk map

Heat map = $G_{\sigma} * M_t$ where M_t is the baseline segmentation mask in standardised crop coordinates and G_{σ} is a Gaussian kernel at $\sigma = 2.5$ voxels. The kernel parameter was selected on a held-out UCSF development subset (N=80) not used in any external validation, and frozen before all reported experiments.

5.2 Lightweight 3D U-Net

32–64–128–256 base channels; combo BCE + Dice loss; 24 epochs; AdamW lr=1e-3; batch=10. 16×48×48 voxel crops. Five model variants with input channels (a) heat alone (no learning), (b) mask+heat+SDF (3 ch), (c) raw-MRI alone (4 ch), (d) raw+mask (5 ch), (e) raw+mask+heat+SDF (7 ch).

5.3 Stronger residual U-Net (v81)

GroupNorm, dropout, source-validation early stopping, source-only affine calibration, H/W-flip test-time augmentation. 18 epochs with patience-3 early stopping. Two seeds (8101, 8102).

5.4 Transformer baselines (v85)

UNETR (Hatamizadeh et al. 2022): feature_size=12, hidden_size=192, mlp_dim=384, num_heads=6, dropout=0.1. SwinUNETR (Tang et al. 2023): feature_size=12. Both at 16×48×48 input size, 22 epochs, AdamW lr=5e-4, batch=4.

5.5 Closed-form crossover and identifiability conditions

Theorem (closed-form crossover under label shift; Saelens et al. 2002 generalised to model-pair). Given two models m_1, m_2 and per-stratum mean Brier $L_{m(c)}$, the unique aggregate-Brier crossover π^* exists if and only if both identifiability conditions hold: C1 ($L_{m_1}(\text{stable}) < L_{m_2}(\text{stable})$) and C2 ($L_{m_1}(\text{active}) > L_{m_2}(\text{active})$). When both hold,

$$\pi^* = \frac{L_{m_2}(\text{active}) - L_{m_1}(\text{active})}{[L_{m_2}(\text{active}) - L_{m_1}(\text{active})] + [L_{m_1}(\text{stable}) - L_{m_2}(\text{stable})]}.$$

For our heat-vs-mask-feature pair on UCSF: C1 (0.041 < 0.140 ✓), C2 (0.274 > 0.199 ✓), giving $\pi^* = 0.43$.

5.6 Statistical analysis

Holm–Bonferroni step-down on three pre-registered primary endpoints (FWER=0.05). All Brier scores: mean +/- SE from 1,000-bootstrap. All p-values two-sided unless stated. Cluster bootstrap for repeated-measures metrics. Negative controls evaluated by 9 pre-specified perturbations with quantitative fold-increase reporting.

5.7 Software, hardware, reproducibility

Python 3.11.9; PyTorch 2.12 (CUDA 12.8); MONAI 1.5.2 (UNETR + SwinUNETR); nibabel 5.4.2; NumPy; SciPy; statsmodels (DerSimonian–Laird). Hardware: NVIDIA RTX 5070 Laptop GPU (8.5 GB VRAM); Intel Core i7 CPU. All scripts versioned at `scripts/`. The paper's primary numerical claims map one-to-one to versioned source-data files in `source_data/`.

CRedit author contributions

[Author roles to be filled at acceptance.]

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI

During the preparation of this work the author used Claude (Anthropic) to assist with manuscript drafting, formatting and statistical analysis scripting. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data and code availability

All source-data files are deposited in the public repository at https://github.com/kamrul0405/Nature_MI_paper, and a frozen Zenodo DOI mirror will be deposited at

acceptance. Public datasets: UCSD-PTGBM (TCIA collection UCSD-PTGBM, DOI 10.7937/fwv2-dt74, CC BY 4.0); MU-Glioma-Post (TCIA collection MU-Glioma-Post, CC BY 4.0); UPENN-GBM (TCIA, CC BY 4.0); LUMIERE (Figshare 10.6084/m9.figshare.c.5904905, CC BY 4.0); PROTEAS-brain-mets (Zenodo 10.5281/zenodo.17253793). Cohorts containing clinical patient data (UCSF-POSTOP, RHHU-GBM, MU-Glioma-Post, Yale-Brain-Mets-Longitudinal) are available from the respective institutions under data-use agreements; requests to the corresponding author.

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Figure captions

Figure 1. Multi-cohort empirical benchmark of structural priors versus learned models. (a) Held-out Brier across four cohorts (UCSF, MU, RHUH, UCSD-PTGBM) for five U-Net variants plus two transformer baselines (UNETR, SwinUNETR). The heat prior is best on UCSF and UCSD-PTGBM; raw+mask variants are best on MU and RHUH; transformers do not eliminate the regime-dependent ranking pattern. (b) Raw+mask – heat Brier deltas with three-seed $\times$ two-architecture bootstrap intervals — every directional outcome preserved across 5 seeds $\times$ 2 architectures $\times$ 4 cohorts (20/20). (c) Held-out winning Brier vs held-out stable-endpoint fraction π_{stable} . UCSD-PTGBM is the documented multi-axis counterexample to a π -only explanation. Closed-form crossover threshold $\pi^* = 0.43$ overlaid. Source image: `figures/main/v78_NMI_raw_loco_stress.png`. Source data: `source_data/v78_nmi_raw_loco_source_data.csv`.